

Josh Sorenson

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PSI Hybrid Rocket

Utah State University Propulsion Research Laboratory

About:

NASA awarded the Utah State University Propulsion Research Laboratory a grant to develop a hybrid rocket motor that simulates lunar landing thrusters. The propellants chosen were gaseous oxygen and ABS plastic. The motor was rigorously tested and iteratively optimized for performance. The final configuration completed more than 100 tests, including 70 hot-fire tests at NASA Marshall Space Flight Center. In 2026 the motor will be installed in the NASA Langley vacuum chamber with lunar regolith simulant to collect plume-surface interaction data.

Objectives and Outcome:

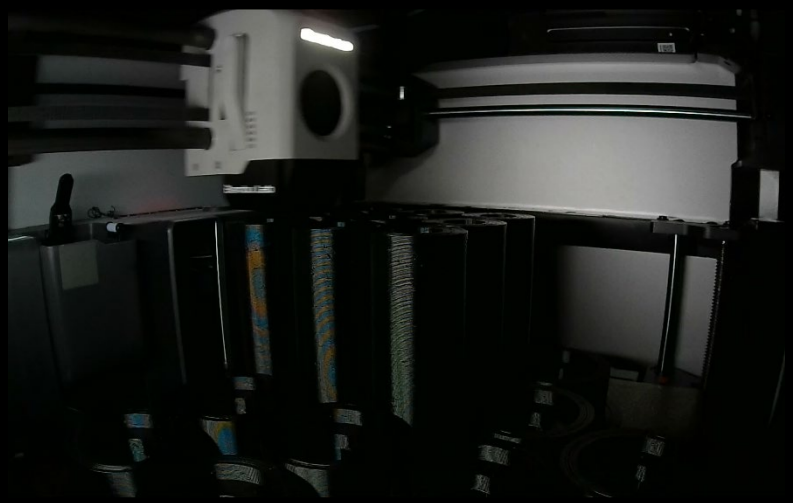
I joined the program after the initial design and prototype hot-firing were complete. My contributions focused on motor performance improvements, hot-fire testing support, and test-stand enhancements. To tune the regression rate I modified the fuel-grain geometry; because the grains were 3D-printed ABS, design iterations were rapid and optimal port sizing was achieved quickly. I redesigned the tool cart and testing workflow to eliminate setup and teardown bottlenecks, cutting total testing time in half. On the test stand I eliminated leaks and added shielding and grounds to cables to reduce data noise. I also served as test conductor for numerous hot-fire tests and processed data from several tests to verify the results of other students.



Testing at the USU PRL



Final Motor Design



3D Printed Fuel Grains



Vacuum Testing at NASA Marshall

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Thesis Topic

Utah State University Propulsion Research Laboratory

About:

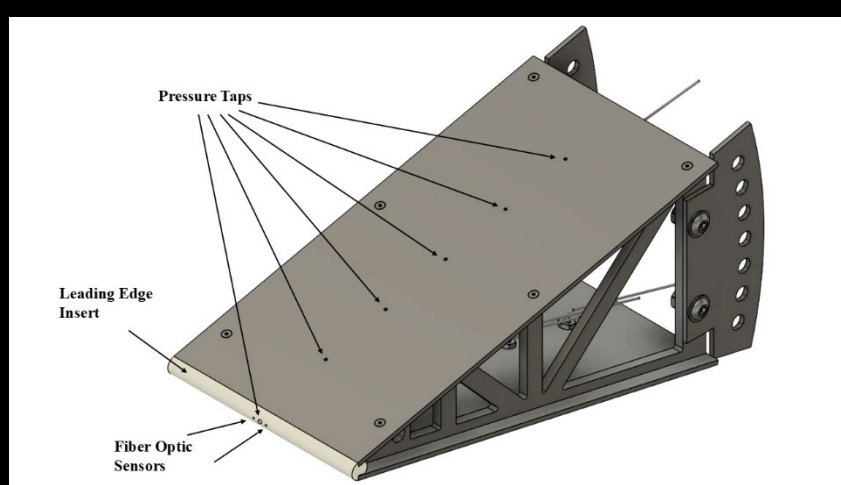
In the field of hypersonic testing, there is a market gap between simulations and test facilities. My thesis topic seeks to fill that gap by using a low-cost hybrid rocket as a high-enthalpy gas generator to simulate hypersonic conditions. To adjust plume temperatures, nitrogen is injected into the plume at the nozzle exit. A sample test article is placed downstream of the plume at a specified distance and is fitted with temperature and pressure ports to obtain stagnation temperature, stagnation pressure, and downstream pressure at specified locations on the article.

Objectives and Outcome:

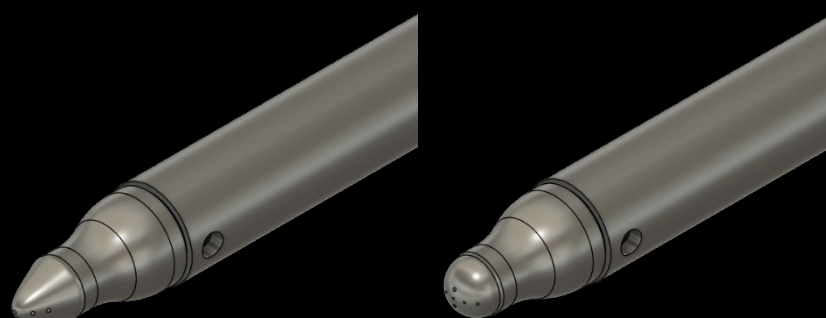
My work on this project centered on improving the technology readiness level (TRL) beyond an initial validation attempt conducted before I began my thesis. Because plume temperatures exceeded the extrapolated Mach number range of interest, I redesigned the test article mounting system and developed a nitrogen injection system to cool the plume. I conducted initial tests without a test article to refine the test process and data acquisition system, then incorporated a previously used wedge test article for validation. Recognizing that an axisymmetric test article would better suit the axisymmetric gas generator, I designed and built two blunt-body test articles and successfully tested both designs. I also machined graphite nozzles to support the testing and developed data processing code for future work with this technology in the Propulsion Research Laboratory. The work was highlighted in a paper that I presented at SciTech 2026, which is now published in the conference proceedings.



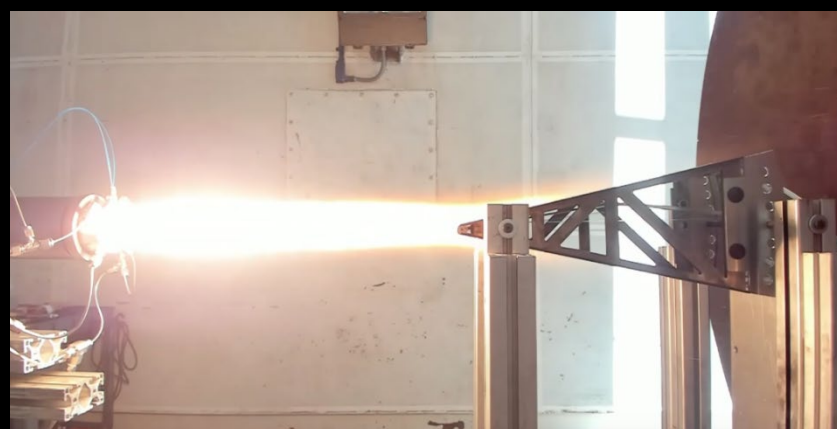
Test Setup



Wedge Test Article



Blunt Body Test Articles



Hot Fire Test on Wedge Article

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Hybrid Rocket Motors

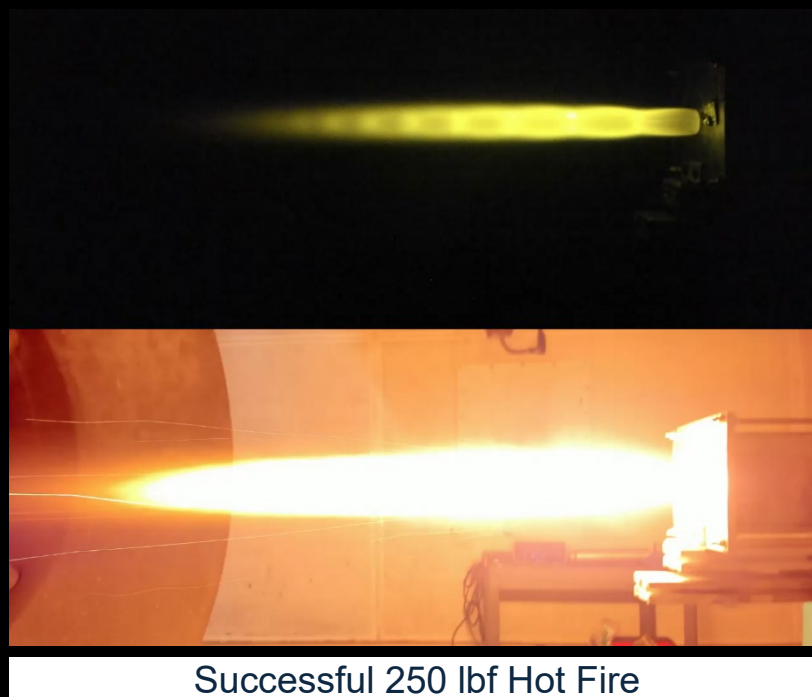
Utah State University Propulsion Research Laboratory

About:

Hybird Space Systems awarded the Utah State University Propulsion Research Laboratory two grants to develop hybrid rocket motors. The first grant focused on a strap-on deorbit motor capable of deorbiting a satellite at end-of-mission life using a nitrous oxide / gaseous oxygen blend with ABS fuel; extensive testing on a work-horse motor led to an optimized injector, fuel grain geometry, and ground-test nozzle such that five perigee burns will successfully deorbit the target satellite. The second grant developed scalable motors targeting 1000 lbf thrust, producing optimized units at 100 lbf, 250 lbf, and 325 lbf scales before Hybird pivoted the effort toward nitrous oxide direct ignition. Both programs utilized the same test infrastructure and shared many development activities.

Objectives and Outcome:

I supported both programs through extensive hardware fabrication, testing, and operations. I machined nozzles and other components on the lathe and mill, procured stock material, and helped manufacture the scaled motors. For both motors I modified fuel grain and nozzle geometries to tune chamber pressure and regression rate. I assembled and disassembled the motors for more than 25 combined hot-fire tests and served as test conductor, running the hot-fire sequence and data acquisition for over 20 tests. I also handled the nitrous oxide blend, manufacturing it prior to each test, and participated in weekly status meetings with Hybird Space Systems. The deorbit motor test stand mounting point was modified for rapid installation and removal, while the scalable motors benefited from stock-part procurement and geometry iterations that enabled rapid thrust scaling up to 325 lbf.



Successful 250 lbf Hot Fire



Successful Deorbit Motor Hot Fire



Spent Deorbit Motor Fuel Grain

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Test Stand

Hybird Space Systems

About:

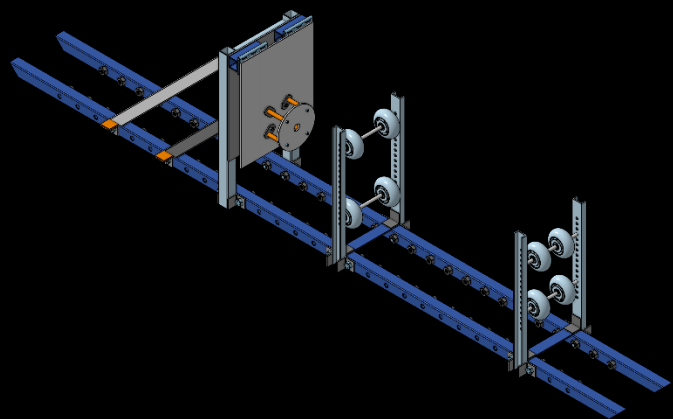
Hybird Space Systems required a 45-degree test stand capable of hot-firing motors up to 10,000 lbf. The stand incorporated a compression load cell mounted on a backplate with a swing plate for rocket motor attachment, which compresses during firing. Designed to support an entire system including oxidizer tank and solid fuel motor, the rocket is strapped to the swing plate and supported by rollers to allow small displacements due to thrust. The stand was completed and has successfully supported 1,000 lbf and 5,000 lbf testing.

Objectives and Outcome:

I performed structural calculations and analysis to ensure the test stand met load capacity requirements and welded the stand. I designed a load-bearing load cell structure that could be adjusted based on rocket length and designed the roller stabilizing structure. During fabrication I resolved challenges with stock material compatibility. I also added a steel plate to protect the concrete where the plume impinges. After installation at the test site, I oversaw mounting procedures for installing motors and added a winch to assist in the mounting process.



Installed Test Stand



Adjustable Motor Mounting Units



Cutting Material



Additional Blast Protection



Hot Fire Test Setup

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Test Cart

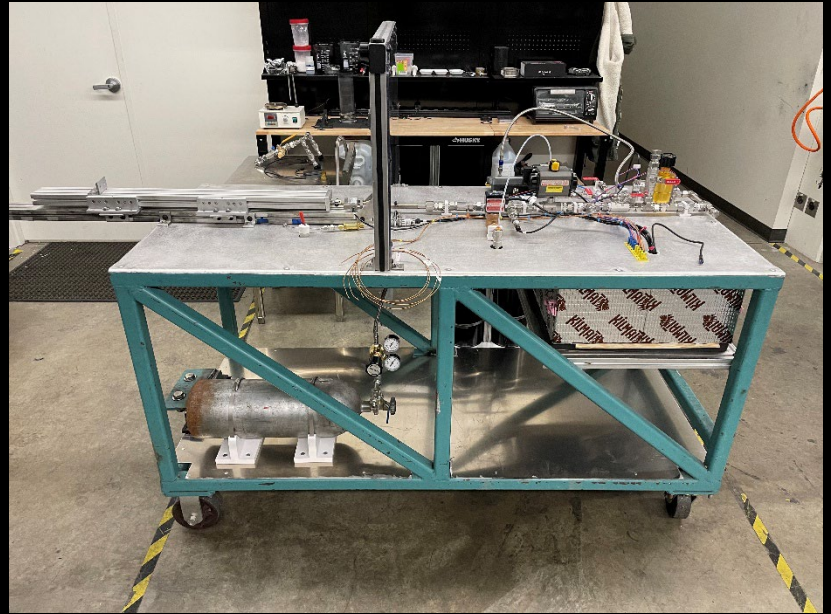
Utah State University Propulsion Research Laboratory

About:

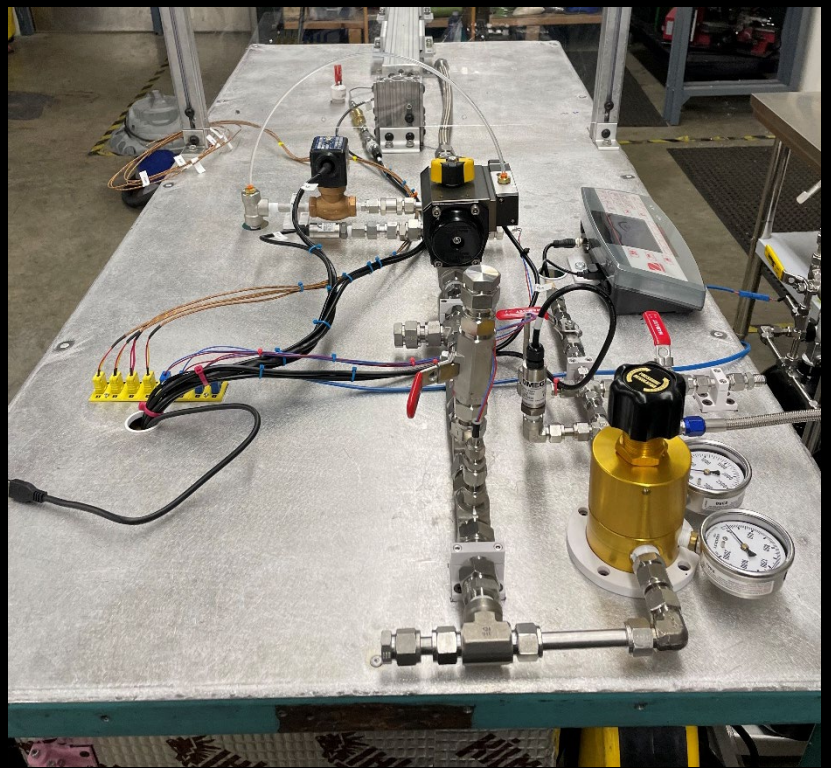
To support the two Hybrid Space Systems grants that were awarded, a new test stand was built atop a cart that could be wheeled into the test cell for testing. An electronics box, oxidizer plumbing, load cell thrust stand, instrumentation, and shielding were all incorporated onto the cart.

Objectives and Outcome:

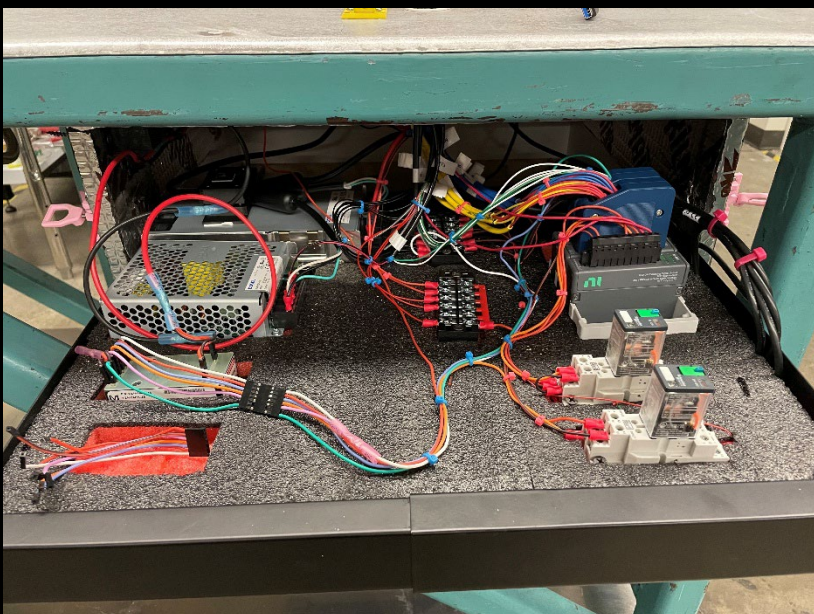
Starting with a scrapped cart, I cut and ground off unnecessary sections to accommodate required components and modified the body to hold oxidizer plumbing, an isolated instrumentation box, and gas tanks. I designed the oxidizer flow path from tank to injector bulkhead, incorporating thermocouples, pressure transducers, and swappable tubing caps for future fittings. I added a mounting point to secure the cart inside the test cell. During testing I fixed leaks, replaced an overloaded load cell, and adjusted the oxidizer plumbing for different motors. The cart has supported over 60 hot-fire tests with thrust capability up to 1000 lbf.



Completed Test Cart



Oxidizer Plumbing Path



Isolated Instrumentation Box



Thrust Stand Sliders

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Thermal Straps

Space Dynamics Laboratory

About:

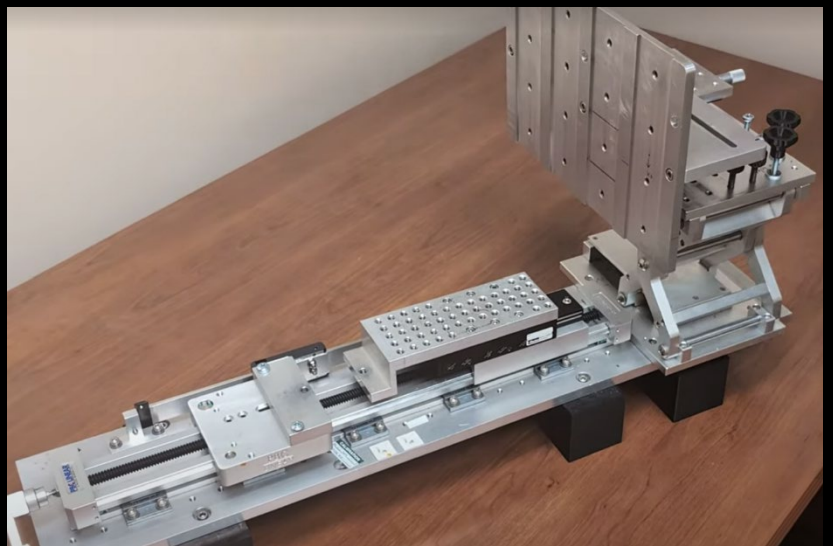
The Thermal Straps group at Space Dynamics Laboratory designs and manufactures flexible thermal links in-house for satellite, telescope, and missile applications. All straps are produced from customer drawings through final packaging for spaceflight integration, including precision machining of pyrolytic graphite, aluminum, or copper foils, cleanroom assembly, multi-axis stiffness testing, vibrational testing, vacuum thermal cycling, conductance testing, precision cleaning, and shipment packaging.

Objectives and Outcome:

I supported five complete programs from kickoff to delivery. For each program I reviewed customer drawings and solid models, then designed test fixtures, acquired fasteners and stock material, and prepared travelers for manufacturing. All programs used aluminum end blocks with graphite-foil thermal paths. I cut foils to length from rolled stock and machined them to exact width on a mill; for two programs I also fabricated the custom fixtures required to mill the foils. I automated the foil cutting process for PGS foils, which drastically reduced lead times for deliverable straps. After assembly and final machining, I ran the stiffness, conductance, and thermal cycling tests using the custom fixtures I designed. Following testing I analyzed the data, prepared customer reports, precision cleaned parts to the required cleanliness level (per UV light inspection and non-volatile residue testing), and packaged the straps in precision-cleaned, custom-designed fixtures that I also created before shipping them with a complete electronic package of reports and deliverables.



Thermal Strap



Stiffness Tester



Vibrational Table



Thermal Vacuum Chamber

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Capstone Project

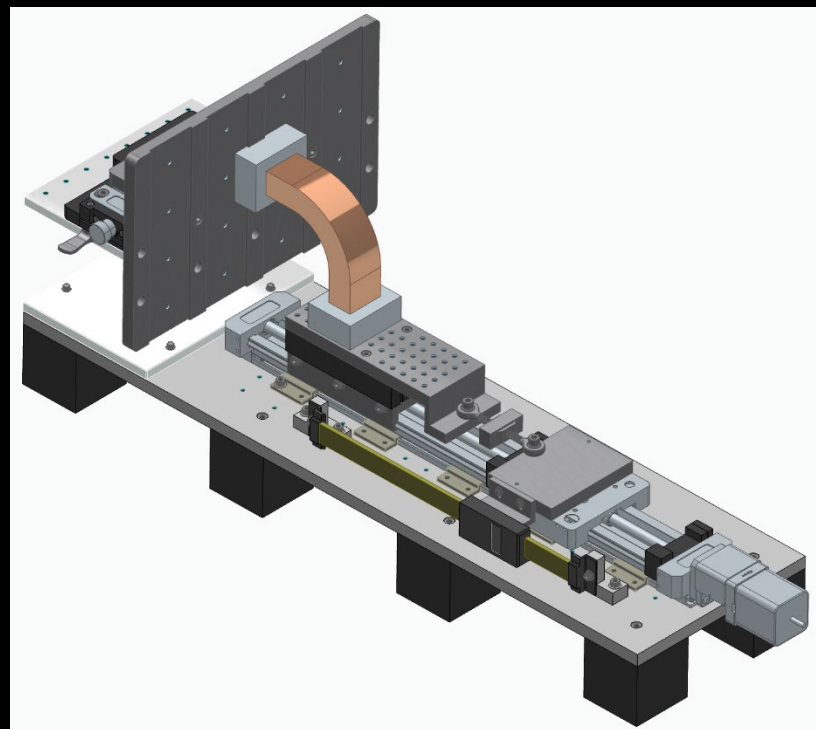
Utah State University

About:

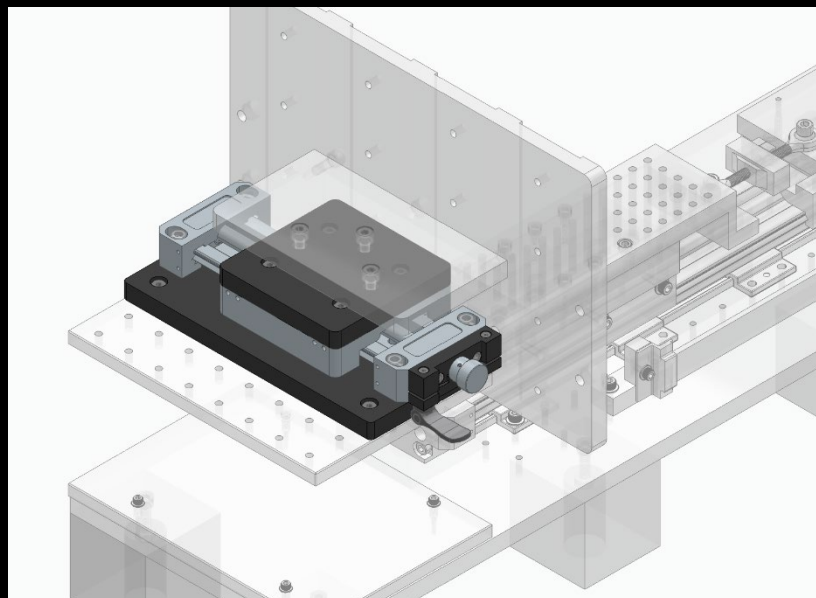
The Space Dynamics Laboratory sponsored a capstone project to upgrade the stiffness tester used for testing the stiffness of straps, making the testing process more efficient. The requested upgrades included a motorized axis that the strap travels along during testing, a lead-screw-driven “y-axis” to replace previous manual screw adjustments, and the construction of a second stiffness tester.

Objectives and Outcome:

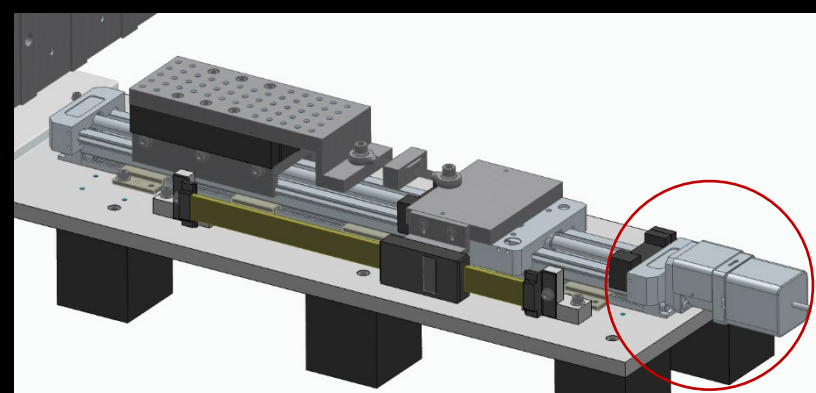
I managed the project timeline to keep the team on track, created the initial budget plan and updated it throughout the program, and sourced materials while submitting purchase orders. I designed the entire stiffness tester in CAD, incorporating all requested upgrades, and created simulations and presentations to convey the design to the project manager. I coordinated with the machine shop to ensure timely part fabrication. With another team member I assembled the new stiffness tester and upgrades. When the company supplying the automated parts delivered the wrong adapter, I contacted them directly and arranged for the correct part to be sent. I built and tested the completed stiffness tester with a test strap to confirm that recorded stiffness values matched between the new and old testers. I also facilitated weekly team meetings and customer reviews to monitor progress and define next steps and contributed to the final report.



Stiffness Tester Solid Model



Y-axis lead screw upgrade



Motorized translational axis upgrade